

Detailed 2-Day Plan by Group

Group A — Person 1 & 2 (Ledoit-Wolf / Part 1)

Day 1 — 27th Feb

Morning (9am–1pm)

Person 1 starts immediately with data. The goal is to have a clean, shared DataFrame ready by late morning so Person 2 isn't blocked.

Person 1:

- Fetch 5Y adjusted close for all 12 tickers using `yfinance`
- Compute monthly returns (`pct_change` on month-end resampled prices)
- Split: first 4Y = train, last 1Y = test
- Save as `monthly_returns_train.csv` and `monthly_returns_test.csv`
- **Target: done by 11am**

Person 2 (while waiting):

- Set up the optimization scaffold — write the Max Sharpe function that takes *any* covariance matrix as input (so it's plug-and-play once Person 1 hands off data)
- Familiarize with `sklearn.covariance.LedoitWolf` API
- **Target: scaffold ready by 11am**

Handoff at 11am — Person 1 shares the returns CSV with Person 2.

Late Morning → Afternoon (11am–5pm)

Person 1:

- Compute Sample Covariance Matrix on train data
- Compute Condition Number $\kappa = \text{np.linalg.cond}(S)$ — document and interpret (anything >1000 is ill-conditioned)
- Write a short bootstrap loop (say 100 resamples) to show how wildly the sample covariance fluctuates — this supports Task 1.1 narrative
- Start building **Viz 1**: side-by-side heatmap using `seaborn.heatmap`, one for sample covariance, one for shrunk (Person 2 will pass the shrunk matrix back)

Person 2:

- Fit `LedoitWolf()` on train monthly returns → get shrunk covariance matrix
- Run Max Sharpe optimization twice — once with sample covariance (from Person 1), once with shrunk covariance
- Extract and compare weights for all 12 stocks
- Identify which stocks show the most dramatic weight shifts between the two
- Identify a "stress window" in the test year (e.g., a period with high volatility) to use as the "Banking Crisis" window for Viz 3

Mid-afternoon sync (around 3–4pm): Person 2 passes the shrunk covariance matrix back to Person 1 so Viz 1 can be completed.

Evening (5pm–9pm)

Person 1:

- Finalize **Viz 1** with both matrices plotted side by side, annotated with condition numbers
- Clean up and comment all Task 1.1 code

Person 2:

- Build **Viz 3**: bar chart or line plot showing MPT vs Ledoit-Wolf weights, specifically highlighting the stressed window — show how LW weights are more stable
- Clean up and comment all Task 1.2 code

End of Day 1 Goal for Group A: Viz 1 done, Viz 3 done, all numbers (κ , weights, shrinkage intensity) computed and verified.

Day 2 — 28th Feb

Morning (9am–12pm)

Person 1 and Person 2 together:

- Cross-check each other's code — verify that the covariance matrices are being computed correctly, optimization constraints are right (weights sum to 1, no shorting if required)
- Verify condition number is being interpreted correctly
- Make sure Viz 1 and Viz 3 are clean, labeled, and presentation-ready

Afternoon (12pm–4pm)

Person 2 leads, Person 1 supports:

- Draft the **Reflection paragraph** for Group A's contribution — focus on why shrinkage is necessary, what the condition number tells us, and how LW improves real-world stability
- Person 1 writes the figure captions for Viz 1 and Viz 3

Evening (4pm–8pm):

- Hand off all outputs (vizzes, code, reflection draft, numbers) to the report assembly team (Person 1 + Person 4 per the earlier plan)
 - Be available for any clarifications during final report compilation
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Group B — Person 3 & 4 (HRP / Part 2)

Day 1 — 27th Feb

Morning (9am–1pm)

Person 3 starts with data, same urgency as Person 1 in Group A.

Person 3:

- Fetch the same 12 tickers but resample to **daily returns** (no resampling needed, just `pct_change`)
- Train/test split: first 4Y daily = train, last 1Y daily = test
- Compute the **correlation matrix** on train data
- Convert to distance matrix: $D = \sqrt{0.5 * (1 - \text{corr})}$
- Run hierarchical clustering using `scipy.cluster.hierarchy.linkage` (use `ward` or `single` linkage — try both and pick the one that gives cleaner clusters)

- **Target: dendrogram cluster order ready by 12pm**

Person 4 (while waiting):

- Build the MPT baseline (Minimum Variance portfolio) using daily train returns — this needs no input from Person 3
- Write the quasi-diagonalization function for HRP (reorders the covariance matrix according to the cluster leaf order — this is where you'll plug in Person 3's output)
- **Target: MPT weights + quasi-diag function ready by 12pm**

Handoff at 12pm — Person 3 shares the linkage matrix and leaf order with Person 4.

Late Morning → Afternoon (12pm–6pm)

Person 3:

- Build **Viz 2**: the dendrogram using `scipy.dendrogram`, properly labeled with ticker names
- This is the most interpretive part of the assignment — spend real time here. Look at which stocks cluster together and label them with your own interpretation (e.g., "Capital Goods + Metals", "Private Sector Banks", "Defensive Consumption") — **do not just label by GICS sector**
- Look at the distance at which clusters merge to understand tightness of relationships
- Write cluster interpretation notes (2–3 sentences per cluster) for the report

Person 4:

- Plug Person 3's leaf order into the quasi-diagonalization step
- Implement the recursive bisection: split clusters into left/right halves, compute inverse-variance weights within each half, scale recursively
- Apply HRP weights to the test year daily returns
- Do the same with MPT weights on the test year
- Compute **Portfolio Turnover** for both: sum of absolute weight changes each rebalancing period (monthly rebalance is a clean assumption)

Evening (6pm–9pm)

Person 3:

- Finalize and polish **Viz 2** — make sure dendrogram is readable, cluster labels are on it, color-coded if possible

Person 4:

- Build the turnover comparison — a simple table or bar chart showing HRP turnover vs MPT turnover across the test year
- Clean up and comment all HRP bisection code

End of Day 1 Goal for Group B: Viz 2 done, turnover comparison done, HRP weights computed and verified against MPT.

Day 2 — 28th Feb

Morning (9am–12pm)

Person 3 and Person 4 together:

- Cross-check: verify the bisection logic is recursing correctly (a common bug is off-by-one in the cluster splitting)
- Verify turnover numbers make intuitive sense — HRP should show lower or comparable turnover to MPT

- Make sure Viz 2 cluster labels are intellectually defensible, not just aesthetic

Afternoon (12pm–4pm)

Person 3 leads, Person 4 supports:

- Draft the **Reflection paragraph** for Group B's contribution — focus on why topology-aware regularization (HRP) is different from numerical shrinkage (LW), and what the dendrogram reveals about market structure
- Person 4 writes captions for the turnover chart

Evening (4pm–8pm):

- Hand off all outputs to report assembly
- Be available for clarifications

The One Thing Each Person Must Not Drop

Person	Non-negotiable Day 1 output
Person 1	Clean monthly returns CSV by 11am
Person 2	Max Sharpe function scaffold ready before the handoff
Person 3	Cluster leaf order ready by 12pm
Person 4	MPT weights + quasi-diag function ready before the handoff

Everything downstream depends on these four deliverables being on time.